

Basic Principles of Medicine 2

Nervous System and Behavioral Sciences

Chemical Senses DLA: Neurologic Examination of Olfaction and Gustation

Dr Bríana Fahey (UK) <BFahey@sgu.edu> and

Dr Walter Kolbinger (Professor Emeritus)

SGU 2026/01



Learning Objectives

A student should be able to:

- **SOM.CS.I.BPM2.1.NB.4.NSCI.0268** Describe neurological examination of olfaction and gustation.
- **SOM.MK.I.BPM2.1.NB.4.NSCI.0271** Contrast the functional anatomy of ageusia, anosmia, olfactory hallucinations in epileptic seizures.



Neurologic Examination of the Olfactory System

- **Olfactory testing:**

- Not routinely carried out
- Ensure use of well-known, non-noxious stimuli (coffee, cloves, vanilla, soap, etc.)
- Patient asked to occlude one of their nostrils with their finger
- Examiner presents a container with olfactory stimulus
- Repeated on the other side

- **Common causes of anosmia and hyposmia:**

- Allergic rhinitis, common cold, smoking, head trauma



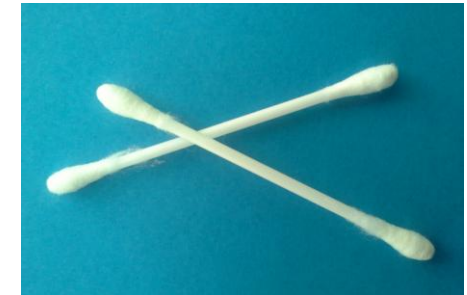
Neurologic Examination of the Gustatory System

- **Gustatory testing:**

- Usually limited to CN VII, innervating the anterior part of the tongue
- Cotton buds dipped in salt solutions, lemon juice for sour, sugar solutions, and tonic water for bitter
- Mouth needs to be rinsed between individual applications
- Each side of the tongue tested separately

- **Common causes of ageusia and hypogeusia:**

- Ageusia is rare, due to redundancy and central divergence of taste pathways. Bell's palsy can result in unilateral ageusia of the anterior tongue.
- Hypogeusia is more frequent. Tobacco use is a common cause.



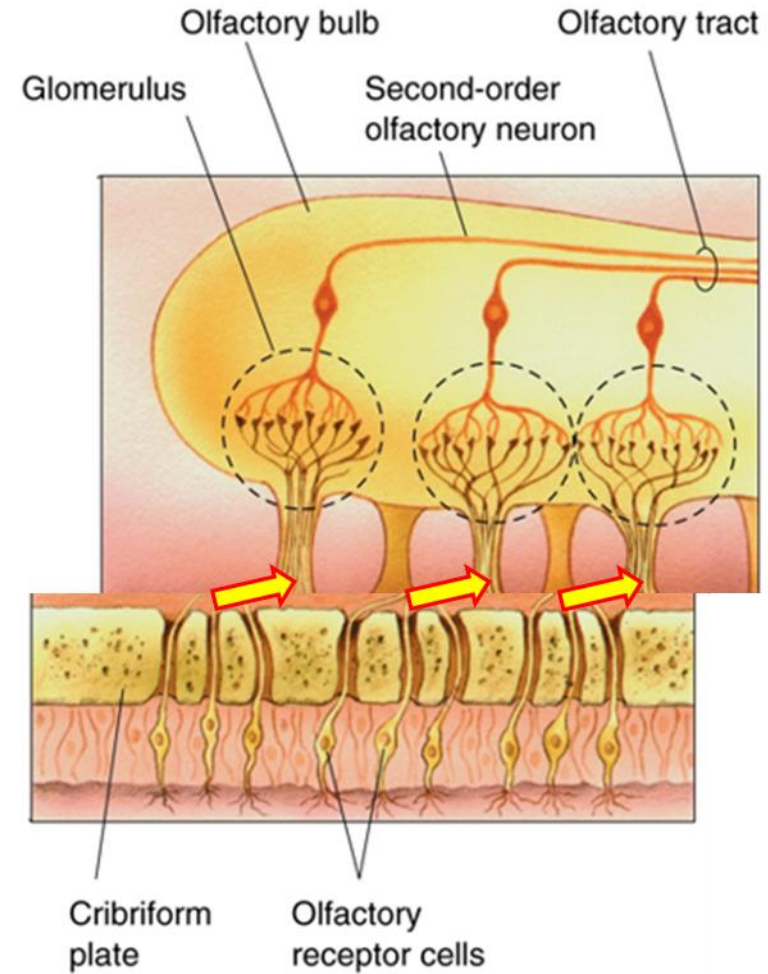
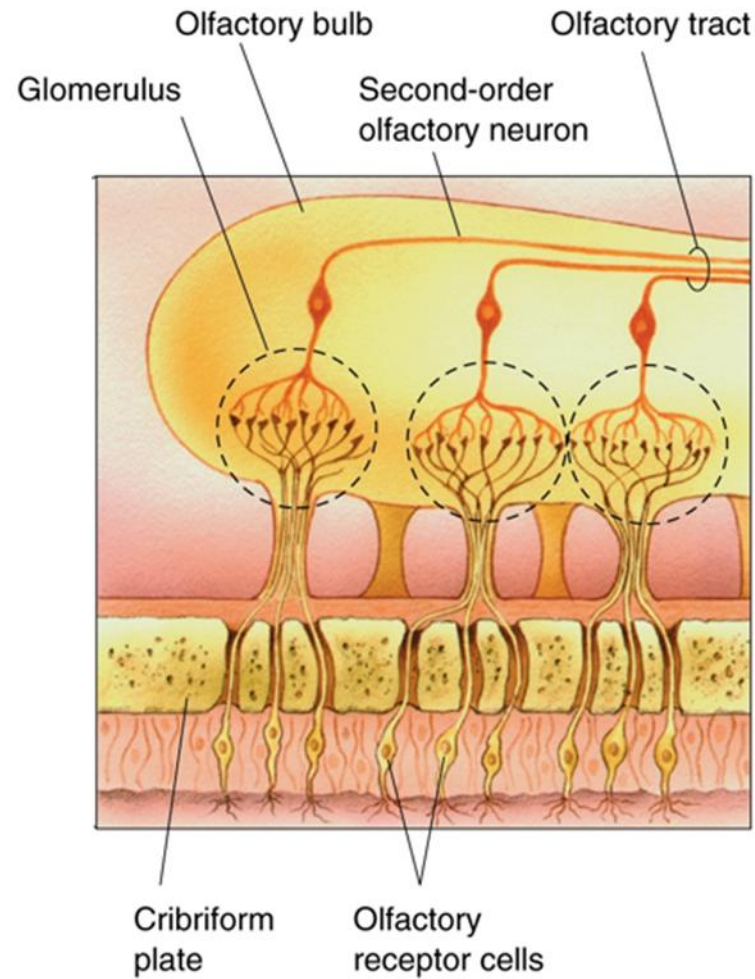
Learning Objectives

A student should be able to:

- **SOM.CS.I.BPM2.1.NB.4.NSCI.0268** Describe neurological examination of olfaction and gustation.
- **SOM.MK.I.BPM2.1.NB.4.NSCI.0271** Contrast the functional anatomy of ageusia, anosmia, olfactory hallucinations in epileptic seizures.



Clinical Correlation – Head Trauma and Anosmia



Clinical Correlations

- Hyposmia (decreased smell function) is four times more common than hypogeusia (decreased taste function), although patients commonly complain they cannot taste, when, in reality, they mean they do not obtain the adequate flavour from food.
- Anosmia (total loss of smell function) may result from the compression of the olfactory tract by tumours, especially meningiomas.
- A patient suffering from epileptic seizures originating in the vicinity of the uncus may sometimes experience olfactory hallucinations, like the experience of a pungent, unpleasant smell, in the absence of odorants that would explain such smell sensations.



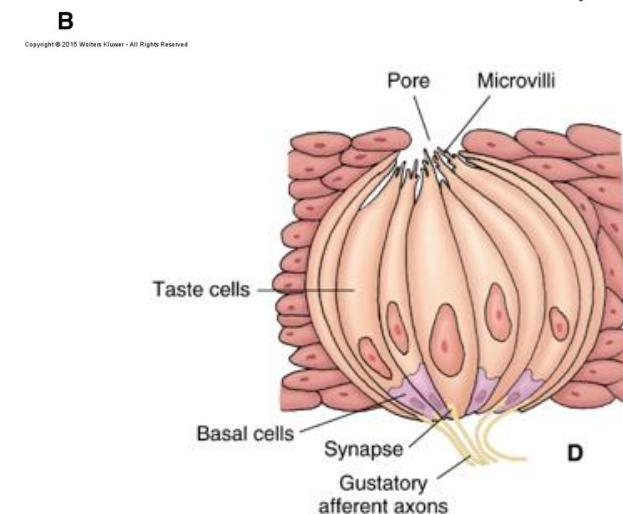
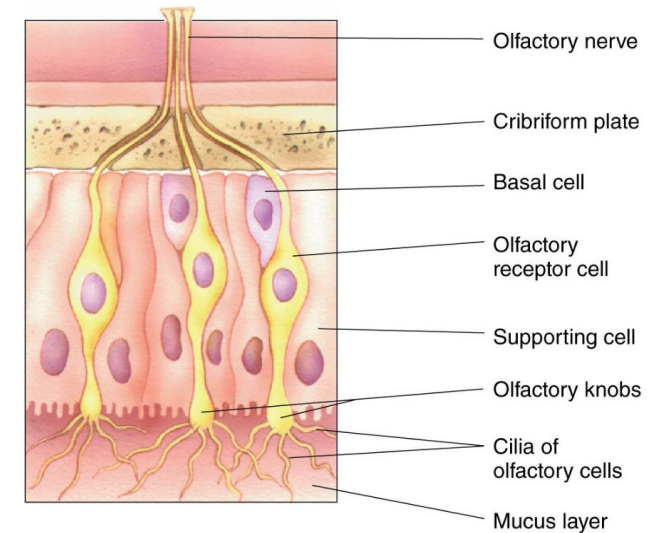
Clinical Correlations

- Dysguesia, a distorted sense of taste, can be a rare complication of **lateral medullary syndrome**.
- This syndrome can arise from occlusion of posterior inferior cerebellar artery (PICA), which supplies the dorsolateral medulla.
- Damage to the part of the gustatory pathway passing through the dorsolateral medulla gives rise to altered taste sensation.



Clinical Correlation – COVID-19 and Chemical Senses

- In the list of clinical symptoms of COVID-19, a sudden loss of sense of smell and taste has been identified.
- This can be found even in the absence of the “usual symptoms” such as fever, cough, respiratory failure.
- A recent study on around 4,000 participants from more than 40 countries confirms that COVID-19 broadly impacts chemosensory function across multiple sensory modalities.
- [Meunier N et al \(2021\)](#) review smell and taste and present a hypothesis for anosmia and ageusia in COVID-19.



The End of Chemical Senses DLA: Neurologic Examination of Olfaction and Gustation

If you have any questions, please send an email to:

Dr Briana Fahey (UK) <BFahey@sgu.edu> or
Dr Noah Leton (GD) <NLeton@sgu.edu>